

Network Routing and Forwarding

draw me a network diagram with routers and end devices and how many routing and forwarding is happening there

Sure! Let's visualize and explain a **simple network diagram** that shows routers, end devices, and how **routing** and **forwarding** work.

Network Diagram (ASCII Art)

less

```
[PC1]    [PC2]
 |        |
 ---      ---
 | R1 |-----| R2 |
 ---      ---
 |        |
 [PC3]    [PC4]
```

Legend:

- PC1, PC2, PC3, PC4 = End Devices (like computers)
- R1, R2 = Routers

How routing and forwarding happen here:

1. Routing:

- Routers (R1, R2) build a **routing table** using routing protocols (like OSPF, RIP, etc.).
- Routing = *Deciding the best path* for data to travel from source to destination.

2. Forwarding:

- When a packet arrives at a router, it looks at the **destination IP** and checks its **forwarding table** (which is usually based on the routing table).
- Forwarding = *Sending the packet to the next hop/interface* based on the table.

Example: PC1 sends data to PC4

- **Step 1:**
PC1 → R1 (directly connected, so no routing here, just local delivery)
- **Step 2:**
R1 receives the packet:
 - R1 checks its forwarding table for the destination (PC4).
 - R1 sees the best path is via R2, so **forwarding** happens from R1 to R2.
- **Step 3:**
R2 receives the packet:
 - R2 checks its forwarding table for the destination (PC4).
 - R2 sees PC4 is directly connected, so **forwards** it to PC4.

Routing happened:

- When R1 and R2 built their routing/forwarding tables.

Forwarding happened:

- R1 → R2 (forwarding)
- R2 → PC4 (forwarding)

How many routing and forwarding events?

- **Routing:**
 - Each router does routing once to build tables (not every packet).
 - **Forwarding:**
 - For every packet from PC1 to PC4:
 - R1 forwards it to R2 (**1st forwarding**)
 - R2 forwards it to PC4 (**2nd forwarding**)
-

Quick Diagram with Forwarding Paths

yaml

```
PC1
|
| (forwarding)
R1 -----\
|          \
| (forwarding) \
PC3          R2
              |
              | (forwarding)
              PC4
```

- **Total Forwarding events for each packet:** 2 ($R1 \rightarrow R2$, $R2 \rightarrow PC4$)

Visual Summary

Routing: Happens in the background, routers calculate best paths and build tables.

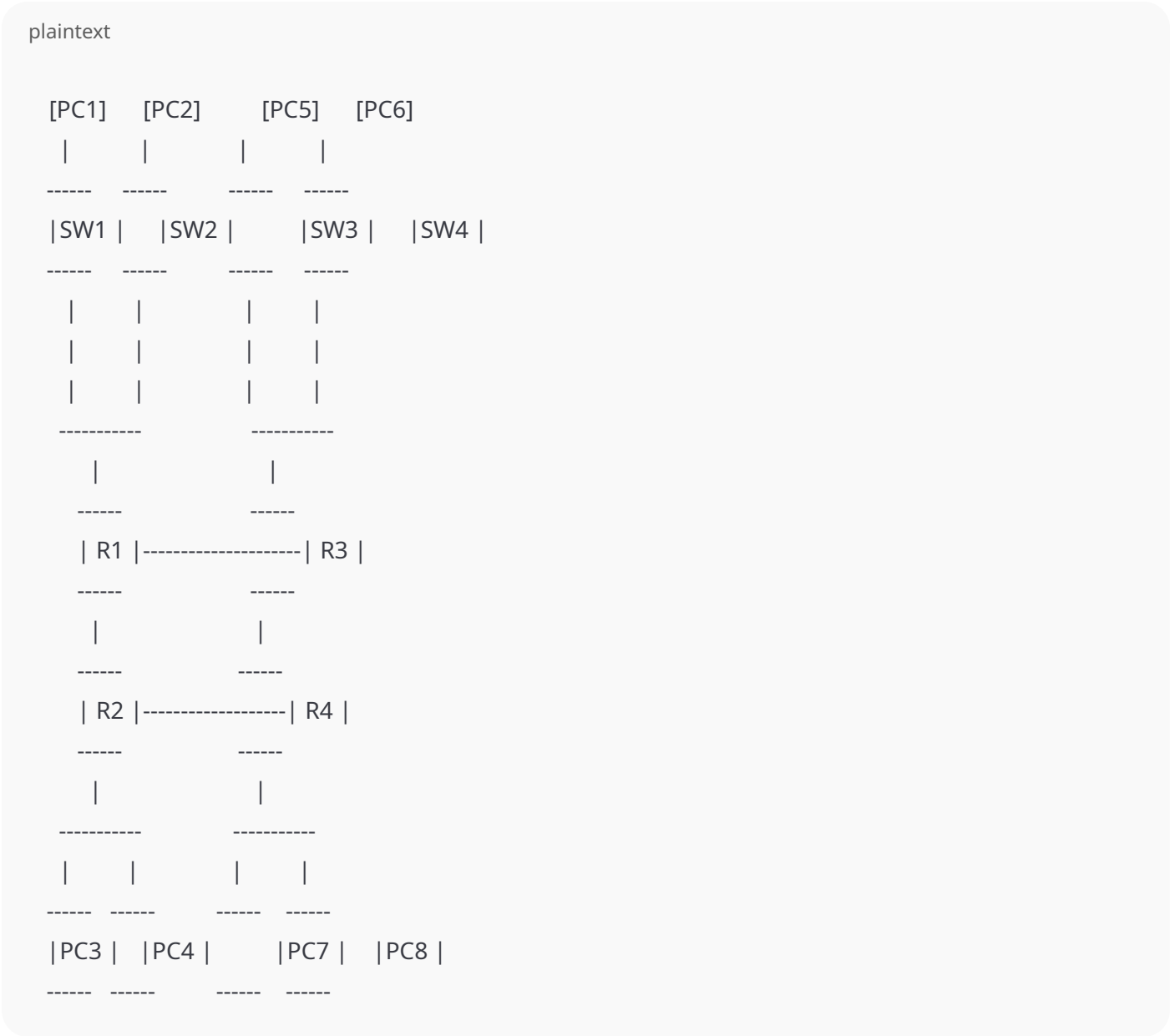
Forwarding: Happens for every packet as it passes through routers.

If you want a more complex diagram or a hand-drawn-style image, let me know and I'll generate a picture version!

more complicated one please

Absolutely! Here's a **more complicated network diagram** with multiple routers, switches, and end devices. I'll also clearly indicate how **routing** and **forwarding** happen at each step with a scenario example.

Network Diagram (Medium Complexity)



Legend:

- PC1-PC8 : End Devices (Computers)
- SW1-SW4 : Switches
- R1-R4 : Routers

Scenario: PC1 wants to send data to PC8

Let's break down how many times routing and forwarding happen in this path:

Step-by-Step Path:

1. PC1 → SW1:

- This is a simple Ethernet switch forwarding (Layer 2, not routing).

2. SW1 → R1:

- Switch forwards to Router R1 (still Layer 2).

3. R1 (Routing):

- R1 checks its **routing table** to find the best path to PC8 (maybe via R3, then R4).
- Routing decision is made here.
- **Forwarding:** R1 sends the packet to R3.

4. R3 (Routing):

- R3 receives packet, checks routing table for PC8.
- Routing decision is made.
- **Forwarding:** R3 sends the packet to R4.

5. R4 (Routing):

- R4 receives packet, checks routing table, sees PC8 is directly attached.
- Routing decision is made.
- **Forwarding:** R4 sends packet out to SW4.

6. SW4 → PC8:

- Switch forwards to PC8 (Layer 2 again).

Summary Table: Routing and Forwarding

Step	Device	Type	What Happens
1	SW1	Forwarding	L2 forward PC1 → R1
2	R1	Routing/Fwd	Routing table lookup + Fwd
3	R3	Routing/Fwd	Routing table lookup + Fwd
4	R4	Routing/Fwd	Routing table lookup + Fwd

Step	Device	Type	What Happens
5	SW4	Forwarding	L2 forward to PC8

Total Routers doing Routing: 3 (R1, R3, R4)

Total Routers doing Forwarding for this packet: 3 times

Total L2 Switch Forwarding: 2 times (SW1, SW4)

Annotated Diagram with Path

plaintext

```

[PC1]
 |
[SW1]
 |
[R1]-----[R2]
 |       |
[R3]-----[R4]
 |       |
[SW4]  [SW3]
 |       |
[PC8]  [PC7]

```

Packet Path:

PC1 → SW1 → R1 → R3 → R4 → SW4 → PC8

- **Routing happens at R1, R3, R4** (each builds a routing table, uses it for the packet).
- **Forwarding happens at every router for every packet.**
- **Switches do only forwarding (Layer 2, not routing).**

Diagram Explanation

- **Routing:** When packet arrives at R1, R3, and R4, each does a routing table lookup to decide the next hop.

- **Forwarding:** After decision, router forwards the packet to the next device.
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Colorful Visual Guide (for your mind):

1. **End Devices:** PC1 → PC8 (edges of the network)
 2. **Switches:** (local traffic, not IP routing)
 3. **Routers:** (make routing decisions, forward packet between networks)
 4. **Routing Table:** Used to decide which router/interface to use next
 5. **Forwarding Table:** Actually pushes the packet to the right interface
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Routing vs Forwarding—At Each Router

Routing (builds table):

- Once per update/change in the network (NOT for every packet).

Forwarding (actual packet movement):

- Every single packet, every time it hits a router.
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If You Want an Image Diagram

If you want an actual graphical image (not just ASCII), let me know and I'll generate a neat, colored network diagram for you!

Summary Statement for Exams

In a network with multiple routers, when a packet travels from one end device to another on a different subnet, each intermediate router makes a routing decision (routing), then sends the packet to the next hop (forwarding). The process repeats at each router along the path, while switches simply forward within the same local network.

If you want a specific number of routers/switches or want to see how this changes for different topologies (like with redundant links), just ask!

another complex example but without switches please

Great request! Here's a **more complex network diagram** with **only routers** (no switches) and multiple end devices connected directly to router interfaces. Then I'll show how **routing** and **forwarding** happen for a data packet traveling from one end device to another.

Complex Network Diagram (Routers Only)

Let's build a scenario with **5 routers (R1–R5)** and **6 end devices (PC1–PC6)** connected as follows:

less

```
[PC1]
  |
[R1]
 / \
[PC2] [R2]
     / \
    [R3] [R4]
   / \ \
[PC3] [R5] [PC5]
   / \
[PC4] [PC6]
```

Network Connections:

- PC1 connects to R1
- PC2 connects to R1
- R1 connects to R2
- R2 connects to R3 and R4

- R3 connects to PC3 and R5
 - R4 connects to PC5
 - R5 connects to PC4 and PC6
-

Scenario: PC1 sends data to PC6

Step-by-step Routing and Forwarding

1. PC1 → R1

- Packet sent to default gateway (R1).

2. R1 (Routing/Forwarding)

- Routing: R1 checks its routing table to find the path to PC6 (which is behind R5).
- Forwarding: R1 sends the packet to R2.

3. R2 (Routing/Forwarding)

- Routing: R2 checks routing table for PC6. Best path: via R3, then R5.
- Forwarding: R2 sends the packet to R3.

4. R3 (Routing/Forwarding)

- Routing: R3 checks routing table for PC6. Best path: via R5.
- Forwarding: R3 sends packet to R5.

5. R5 (Routing/Forwarding)

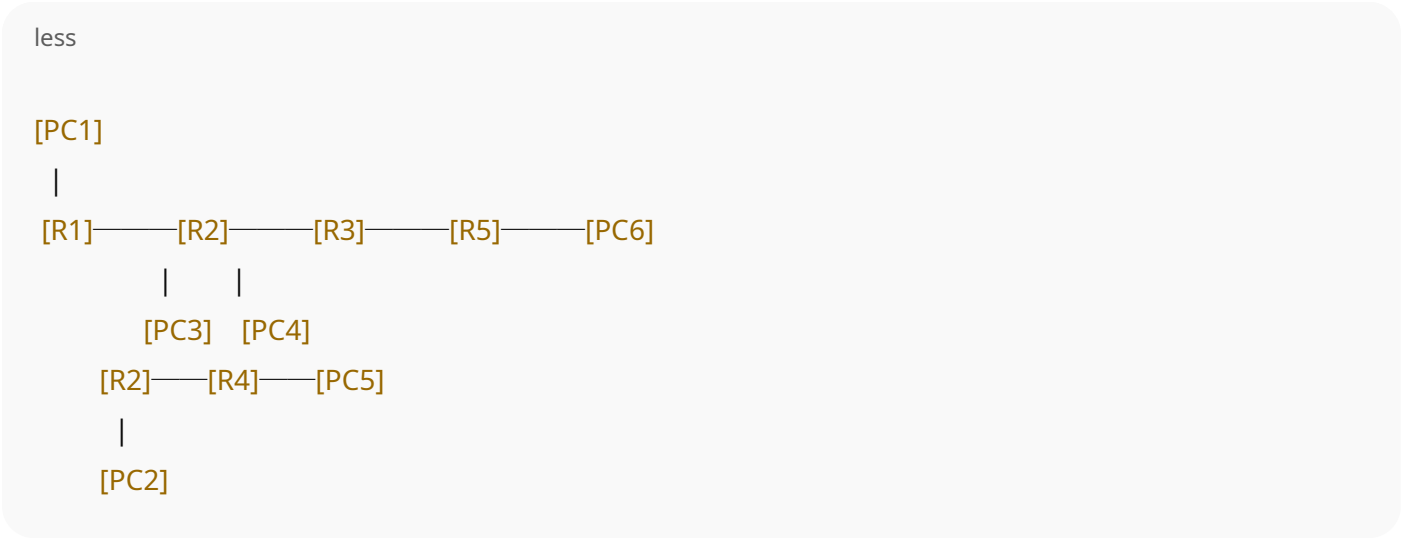
- Routing: R5 checks its interfaces and routing table. Sees PC6 is directly connected.
 - Forwarding: R5 sends packet to PC6.
-

How many times routing and forwarding happen?

- **Routing decisions (table lookup):** At every router (R1 → R2 → R3 → R5)
- **Forwarding:** At every router (same as above, per-packet)

Total routers involved: 4

Visual Path



- **Data path:** PC1 → R1 → R2 → R3 → R5 → PC6

Diagram Key Points

- **No switches:** All connections are direct to router interfaces.
- **Routers do routing and forwarding:** Every router involved checks its routing table (routing) and forwards the packet to the next hop (forwarding).
- **End devices only send to their default gateway (router).**

Summary Table

Step	Device	Routing	Forwarding	Description
1	R1	Yes	Yes	To R2
2	R2	Yes	Yes	To R3
3	R3	Yes	Yes	To R5
4	R5	Yes	Yes	To PC6 (directly connected)

Explanation

- **Routing** happens when the routers decide where to send the packet next, based on the destination IP and their routing tables.
 - **Forwarding** is when the router actually sends the packet out the correct interface.
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If You Want a Visual Image

Let me know if you want a graphical image version—I can generate a clear, colored network diagram for you!

Final Exam-Ready Note

In a multi-router network without switches, each router directly connects to both end devices and other routers. For a packet to travel from one end device to another on a different subnet, it passes through multiple routers. At each hop, the router makes a routing decision (using its routing table) and then forwards the packet out the correct interface. This process repeats until the packet reaches the destination device.

Need an even larger or more branching network, or want to see how routing loops or redundant paths would look? Just ask!